



Outline of the lecture

The history of Informatics & HCI

Our roots and evolution

Conceptual Foundations

Information Processing | Gestalt Theory | Functional Approaches

Action Research & Praxeology

Work(place) studies & ethnography | Contextual Approaches |
Praxeology & Appropriation | Activity Theory

Structuration & Infrastructuring

A holistic look at Human-System-Interaction

HCI in Practice

Designing | Prototyping | Evaluation

Giddens's Theory of Structuration (ST)

"Humans communicate through structures to adopt procedures and activate resources needed to accomplish tasks, goals and resolutions. This gives birth to systems" (Giddens, 1984)



Giddens's Theory of Structuration (ST)

Structuration theory: how social structures are both the medium and the outcome of social interactions



Division of perspectives

Social phenomena can be described as the result of human action.
Human action can be understood by objective, exogenous socio-cultural framework conditions.

Aim: The integration of both perspectives ("The constitution of society", Giddens 1965)



Duality of Structure

Social structures as results and limitations of human actions

Actions create structures

and

Structures enable and limit future actions



Why is it interesting for HCI?

Studies have shown both the enabling and constraining character of information systems.

The production/introduction of information systems as structure-creating action.

Agency is the capacity of individuals to have the power and resources to fulfill their potential. For instance, **structures** consists of those factors of influence (such as social class, religion, gender, ethnicity, ability, customs, etc.) that determine or limit and their decisions.

ST: Structure types and modalities

Basics: A persons rules are created from three basic structures

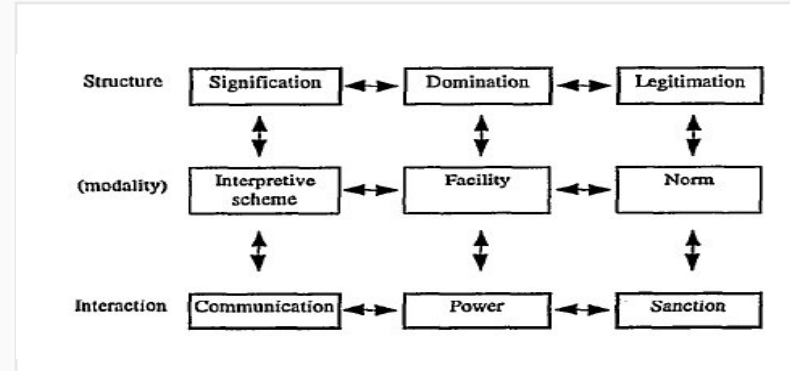
Structure types: not distinctive categories of analysis, mutually influencing. Interaction with agents through three modalities: Interpretation schemata, resource processing facilities and norms

Signification: How an event should be interpreted? Production of meaning through language/character sets (semantic codes, interpretation schemata, discursive practice) e.g. price label in a shop: Detailed description, utility, exchange value; Bank notes: exchange value; Interpretation schema allows the equivalence of values and amount of money to me understood. Regulation by communication.

Domination: What means should be used to accomplish goals? Production of power by controlling resources, e.g. money provides the power to speed up the handing over of goods Regulated by the exercise of power

Legitimation: What should happen in a given situation? Creation of a moral order by embedding in norms, values and standards, e.g. (re)negotiation of the advertised prices common/uncommon depending on cultural values. Regulated by sanctions.

These structures are not static; they exist only as they are enacted in social practices.



Modality: a particular mode in which something exists or is experienced or expressed - the mechanisms through which structure is enacted and mediated in interaction

ST: Structure types and modalities

Basics: A persons rules are created from three basic structures

Signification ↔ **Interpretive Schemes** ↔ **Communication** (How money gets meaning)

This refers to the shared meanings people have in society.

With money, the shared meaning is:

- “This paper/number represents value.”
- “\$10 is worth more than \$5.”

Modality: Interpretive Schemes -> These are the cognitive frames people use to apply meaning.

For money, interpretive schemes include:

- Knowing what a price tag means,
- Understanding what a paycheck signifies,
- Understanding currency symbols

Interaction: Communication -> In daily life, people communicate these meanings:

- A cashier tells you the total price
- A bank statement “tells” you your balance
- People negotiate prices

➔ Money functions because people share and use the same interpretive schemes while communicating about value.

ST: Structure types and modalities

Basics: A persons rules are created from three basic structures

Domination ↔ **Facilities** ↔ **Power** (How money is a resource that gives power)

Domination refers to structures of power and control.

In the case of money:

- Banks, markets, financial systems control resources
- Wealth differences create power differences

Modality: Facilities -> Facilities are the resources people draw upon in action:

- Physical money, credit cards, banking apps
- Central banks and financial institutions
- Economic capital itself

Interaction: Power -> In interactions, people use these resources to exercise influence:

- You can buy things that others cannot
- Employers pay wages and influence employees
- Governments tax citizens and fund programs

➔ Money is a facility that gives individuals and institutions power in everyday interactions.

ST: Structure types and modalities

Basics: A persons rules are created from three basic structures

Legitimation ↔ **Norms** ↔ **Sanction** (How rules about money shape behavior)

These are the rules that define what is legitimate, legal, or expected.

In the money system:

- Laws defining legal tender
- Rules against counterfeiting
- Social norms about paying debts

Modality: Norms -> Norms are the standards people use to judge actions:

- “You must pay for what you buy.”
- “Don’t steal money.”
- “Pay your taxes on time.”

Interaction: Sanction -> In everyday interaction, people enforce or face consequences:

- You are fined for not paying taxes
- You go to jail for counterfeiting
- Your credit score drops if you don’t repay loans
- Social disapproval if you don’t split a bill properly

➔ Norms and sanctions maintain the legitimacy of the money system.

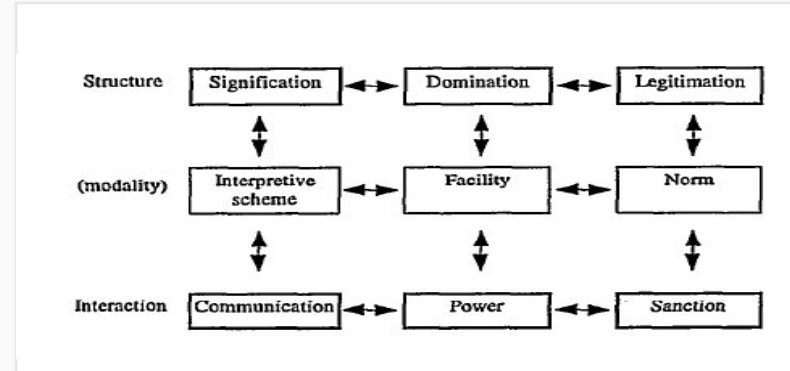
ST: What is “structure”?

Central term and characteristics

Structure is not ‘objective/measurable’ because it exists solely through human actions. Humans are in a state of **reflexive consideration (reflective agency)** regarding their situation and the omnipresent potentials for change. Where is the “structure” in information systems?

- **Duality of Structure:** Structures are both enabling and constraining. They shape interactions while being reproduced through them.
- **Recursive Process:** Structures are not external forces but are continuously created and modified through social practices.
- **Agency and Structure Interdependence:** Individuals (agents) draw on structures during interactions, and their actions simultaneously reproduce or transform these structures.

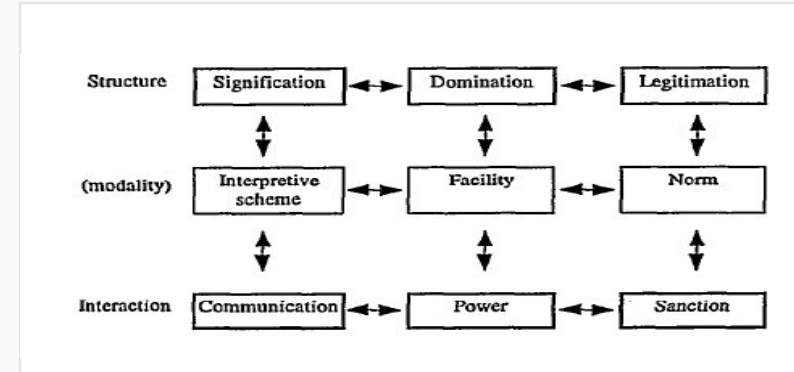
Main points of criticism: Very soft definition of ‘structure’, leaves room for interpretation, Reference to the action (Agency)



ST: What is “structure”?

Central term and characteristics

Definition of structuring: Systems are the reproductive relationships between agents or collectives (organized as social practices). *Structure* is (or rather structural characteristics are) rules and resources which emerge as characteristics from systems. *Structuring* is the entirety of the parameters which influence the continuity or modification of structures and therefore the reproduction of systems (Intricacies of the definition first appear when compared with other sociological theories).



Capability vs Knowledgeability

ST and HCI

Naive approaches

Orlikowski 1992: Structure is embedded in information technology:

Structure emerges from the actions of programmers. Structure is relatively hard wired in information systems. The results of the analyses of the effects of IT: Techno-determinism – the behaviour of users is governed by the embedded structures
Studies e.g. on the introduction and usage of Groupware

Scheepers and Damsgard 1997: Comparison of the structural characteristics of diverse intranets. Result: Recommendations for the set-up of in-house intranet

Table 1: Overview of analysed cases

Structural Dimension	National Semiconductor Corporation	SAS Institute	ORNL/LMES	Bectel Group Inc.
Signification	Initial lack of mutual signification structure for the intranet.	Lack of shared signification structures lead to initial rejection.	WebWeek launched to establish shared structures of signification.	Top management support ensured shared signification structure.
Domination	Management threat to put an end to the intranet implementation.	Management intervene in direct support for the intranet.	The working group realizes that it needs management support.	Domination structures applied to ensure uniform layout and standards.
Legitimization	The champions interact with management acknowledging existing power structures and that intranet technology is in line with organizational values and norms.	Intranet implemented in accordance with social structures and norms.	The intranet fits well with some parts and poses a challenge to legitimization structures in other parts.	Homogenous fit with legitimization structures enabled a smooth intranet implementation.

ST and HCI

Orlikowski (1992, important correction 2000)

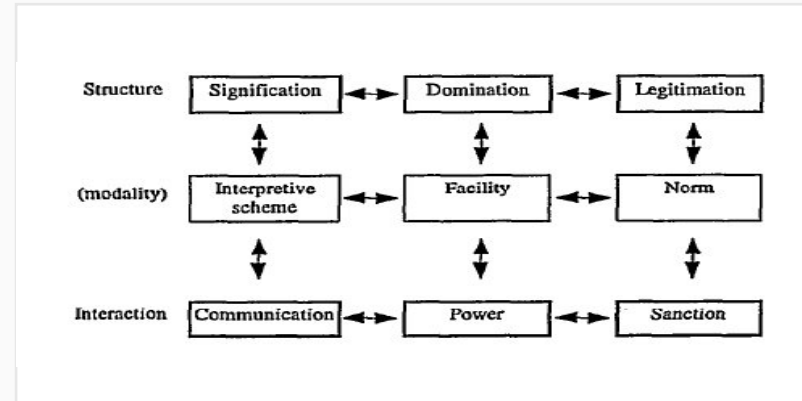
1992: Duality of technology: Technologies as material artefacts but not in focus. Decoupling of activities makes it possible to perceive artefacts as the result of human activity. The interpretation of artefact usage is free but limited by functionality. The institutionalisation of structure.

Correction 2000: Only the **perceived and regularly used characteristics of technology** influence structurisation. Not technology but **the use of technology creates structure**. The use of technology is emergent, not only shaped by designers but also by user activities. Institutionalisation only “for now”.

Methodological execution:

Qualitative empiricism: The use of ethnographical and ethnomethodological methods, focusing on the lines of conflict which are described by the concept of the duality of technology:

- Who is influencing technological structures (by designing/developing them, by configuring them, through not using them)?
- Who is affected by technological structures (by (not) being able to do something)?



ST and HCI

Naive approaches

Structure	Modality (Interpretive Schemes, Facilities, Norms)	Interaction
Signification: Shared understanding of customer data as valuable insights for decision-making. The CRM organizes sales metrics, customer preferences, and engagement statistics into meaningful dashboards.	Interpretive Scheme: Training manuals, tutorials, and shared knowledge about how to use the CRM system (e.g., inputting data, running reports).	Communication: Sales team members log client interactions, and managers extract reports to share sales forecasts with the team.
Domination: Managers use the CRM as a tool to allocate leads, monitor sales activity, and assign resources effectively. The system centralizes power over customer management.	Facility: The CRM system acts as a resource, enabling actions (tracking leads, automating tasks) but also restricting them (e.g., requiring approvals for discounts).	Power: Managers use the CRM to monitor employee performance, reassign leads, and influence team priorities.
Legitimation: Policies require employees to regularly update the CRM and ensure data accuracy. Non-compliance affects performance reviews.	Norm: Employees are expected to log all customer interactions and complete lead updates in the CRM daily.	Sanction: If employees fail to follow the norms (e.g., neglect data entry), they may face performance warnings or reduced bonuses.

ST and HCI

Adaptive Structuration Theory (AST), De Sanctis and Poole (1990, 1994)

Target statement: “Given an innovative information technology and further sources of social structures n_1 to n_k as well as pertinent appropriation structures and task-oriented decision processes, IT is used as intended.”

Method: Analysis of the ‘structural features’ of the technologies and the ‘spirit’. Structural features: e.g. a voting algorithm in a system which supports decision making. *Spirit*: the quantity of values and aims which form the basis for the implementation of ‘structural features’. *Appropriations*: immediately visible activities which testify to underlying structuration processes. ‘*Appropriation moves*’: activities of groups who adopt the ‘structural features’ of the IT-designers’ usage intentions either as intended (‘faithful’) or not (‘unfaithful’).

Criticism: The result is a positivistic framework (the acceptance of an objective cause-effect relationship which need only be empirically proven). Has but little in common with Giddens descriptions (e.g. immateriality of structure), see Jones 1999.

The types of structures that are provided by advanced technologies and the structures that actually emerge in human action as people interact with these technologies define what the use of technology will be



The theory purports that structures in technology and structures in action are continually intertwined, continuously shaping each other.



Group dynamics are too complicated to be reduced a few propositions or a predictable chain of events; group members intentionally adapt rules and resources to accomplish goals

ST and HCI

Adaptive Structuration Theory (AST), De Sanctis and Poole (1990, 1994)

